

The Σ -Model in Protein Domain Architectures: Schema-Coherence Suppression and Recombination

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Abstract

[PLACEHOLDER — 200-word abstract describing the application of the Σ -Model framework to protein domain architecture recombination, introducing the PfamCG-1.0 benchmark, and reporting experimental results for brittle recombination, grammatical induction probes, and curriculum design.]

1 Introduction

[PLACEHOLDER — Motivation: compositional generalisation in protein domain architectures. The Σ -Model [1] as a dynamical-systems framework for schema-coherence suppression. Three contributions: brittle recombination (N=500), grammatical probes (3 probe types), and PfamCG-1.0 benchmark suite.]

2 Background: The Σ -Model Framework

[PLACEHOLDER — Summary of the coupled ODE system (Eqs. 1–6 of Basri, 2026), key concepts (σ_A , δ_A , α_A), phases of learning, and the training-aware curriculum hypothesis.]

3 Pfam Domain Architectures as a Test Bed

[PLACEHOLDER — Pfam database (version 37.0), domain architecture representation, recombination distance via HMM bit scores, vocabulary construction, and dataset statistics.]

4 Experiment 1: Brittle Domain Recombination

[PLACEHOLDER — N=500 regime: familiar combinations, new families, and cross-fold transfer. Hypothesis: Σ -Model predicts brittle recombination for unfamiliar domain pairs. Results: accuracy vs. familiarity, effect size, statistical tests.]

5 Experiment 2: Grammatical Induction Probes

[PLACEHOLDER — Three probe types: positional bias (linear), co-occurrence (bilinear), hierarchy (MLP). Results per probe: accuracy, R^2 , and comparison to random baseline.]

6 Experiment 3: PfamCG-1.0 Benchmark

[PLACEHOLDER — Benchmark results: recombination accuracy, domain retention rate, novel combination precision, fold consistency. Comparison with baselines: random, frequency-based, co-occurrence.]

7 Discussion

[PLACEHOLDER — Implications for protein engineering, limitations (vocabulary size, architecture depth), connections to other compositional generalisation benchmarks.]

8 Related Work

[PLACEHOLDER — Protein language models (ESM, ProtBERT), domain architecture prediction, curriculum learning for biological sequences, compositional generalisation in NLP.]

9 Conclusion

[PLACEHOLDER — Summary of findings and future directions.]

References

[1] Basyirin Amsyar Basri. The σ -model: Schema-coherence suppression as the origin of compositional generalisation failure. *Transactions on Machine Learning Research*, 2026. Forthcoming.

A Supplemental Materials

[PLACEHOLDER — Additional experimental details, hyperparameter grids, and full result tables.]